

Source-verified human variant experiments

25 substitutions · 6 protein study families · 43 protein family-substitution cells

		Protein						RNA
		Font 2001 HeLa	Pineda 2004 HeLa	Shigeta 2006 COS-7*	Bartoccioni 2008 HeLa	Yan 2020 Liposomes	He 2026 HEK293	Gaildrat 2017 Minigene
SLC3A1	L89P				Q			
	V183A					P		
	T216M				Q	P		
	R365W		N/Q		Q			
	M467K				Q			
	M467T				Q	P	C	
	P52L†					P		
SLC7A9	A70V	N				R	N	S
	G105R	N					N	
	T123M					R		
	V142A			Q				
	V170M	N				R	C	
	A182T	N				R	N	
	G195R			Q				
	L223M			Q				
	N227D†			Q				
	W230R†					Q		
	G259R					P		
	R333Q			Q				
	R333W	N		Q		R	N	
	A354T	N				Q	C	
	A382T†					R		
	M477L‡			Q				
	P482L			Q/L		R	C	
	P482S‡			Q				

- N

Published numerical summary
- R

Textual group range
- P

Purification / stability observation
- S

Allele-specific splicing result
- C

Strict upper bound (<5% WT)
- Q

Qualitative transport result
- L

Protein detection / localization

† Outside the strict ClinVar set. ‡ Current protein match from engineered-only evidence.

* Shigeta uptake: COS-7; detection/localization: COS-7, MDCK II and HEK293.

N/Q and Q/L preserve distinct conditions or endpoints. RNA: c.209C>T (A70V).

Blank: no qualifying result in this curated matrix. Evidence does not specify clinical effect.